MUHAMMAD ZAHID	BT201 Ecology, Biodiversity & Evolution-II
MCQs	Fall 2020 (December 2020)
Grand Quiz	Midterm

1. The complete and published genomes of plant by June 2005.....4
2. The complete and published genomes of Apicomplex by June 2005 6
3. The complete and published genomes of Entamoeba by June 2005 1
4. The complete and published genomes of archaea by June 2005..... 21
5. The complete and published genomes of animal by June 2005 16
6. The complete and published genomes of bacteria by June 2005 211
7. The complete and published genomes of aves by June 2005 1
8. The complete and published genomes of Bacillariophyta by June 2005 1
9. The complete and published genomes of Pisces by June 2005 03
10. The complete and published genomes of Mammalia by June 2005. 5
11. The complete and published genomes of Nematoda by June 2005..... 2
12. The complete and published genomes of tunicata by June 2005. 1
13. The complete and published genomes of insecta by June 2005. 4
14. The complete and published genomes of Myxomycota by June 2005. 1
15. The complete and published genomes of Kinetoplastida by June 2005. 2
16. The complete and published genomes of Cryptomonadina by June 2005. 1
17. The complete and published genomes of Bacillariophyta by June 2005. 1
18. The complete and published genomes of Rhodophyta by June 2005. 1
19. The complete and published genomes of Fungi by June 2005. 14
20. The complete and published genomes of Total by June 2005. 279
21. The complete and published genomes of eukarya in total by June 2005..... 47
22. Saccharomyces is the study of all the ..... in the cell.....carbohydrates

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23._____ The elements that can 'jump' into the DNA of drosophila for introduction of foreign DNA…… P

24.Species richness is generally highest in _____ latitudes, and then decreases to the north and south, with zero species in the Arctic regions…….. Tropical

25.Brittle star belongs to which phylum ……Echinoderms

26.Leeches belong to phylum…… Annelida

27.A ..... grouping consists of various species that lack a common ancestor…polyphyletic

28.A.... is a group of species that includes an ancestral species and all its descendants…… clade

29.One of the following option is not an example of Ex situ conservation of species……natural parks and nature reserves

30.the smallest units that can sustain life in isolation from all but atmospheric surroundings is known as…….. Ecosystem

31.Habitat destruction is a major cause for…… biodiversity loss

32.Class Clitellata belongs to phylum…… Annelida

33.The template nucleic acid used for translation is…… mRNA

34._____ uses DNA and other molecular data to determine evolutionary relationships ……Molecular systematics

35.Adult Echinoderms are ……radially symmetry

36.Australopithecus afarensis was first bipedal hominid found in east Africa about……3.0-3.2 MYBP

37.Phanerozoic consists of ……all of them

38.1n an individual organism, is the state of possessing different alleles at a given locus in regard to a given character? …… Heterozygosity

39.The puffer fish, takifugu rubripes one-eighth of ..... genome ……Homo sapiens

40.The kudzu plant was introduced to the..... from Asia in the late 1800s as an ornamental plant ……America

41.Gymnosperms have ……naked seeds

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42. large areas of protected land for conservation of biodiversity and the traditional life of the tribal living there……..**Biosphere reserves**

43. Earliest fossils dated to 200 million years ago are …….. **Turtles and tortoises**

44. RNA shows enzymic (catalytic) properties called …….. **Ribozymes**

45. Genetic diversity, means that the population contains most of the possible alleles for each particular ……**gene locus**

46. Zoologist who study birds are called ……**Ornithologist**

47. Some expert's estimate that around 30% of all species on earth will be extinct by……..**2050**

48. Different individuals carry different genetic information, The sum of the variation of an interbreeding population is called…… **Gene pool**

49. sequencing using mass spectrometry, in which masses are estimated for ..... generated from proteolytic digests of the protein ……..**peptides**

50. how the birds are designed to fly? ……**Three toes pointing forward and one toe pointing backward.**

51. Cyanobacteria introduced significant amount of ……**O_2**

52. Ticks belong to phylum ……**Arthropoda**

53. Arctic communities include just a ..... species packed into each cubic meter……. **few**

54. Ballooning occur in ……**Chelicerata**

55. Which of the following system is found in Echinoderms?... system of internal skeleton

56. ..... species are those species of plants and animals found only in a particular area and not found anywhere else…… **Endemic**

57. Which of the following structure is present in all the chordates…… **Notochord**

58. The protease used in Mass spectrometry is ……**Trypsin**

59. the study of ecology in urban areas and the interactions between organisms is called .... ecology……**urban**

60. One of the following is not the consequences of deforestation……**Decrease in the level of carbon dioxide in the atmosphere**

61. Notochord is ……**flexible**

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62. Spider belong to subphylum ……Chelicerata

63. Human genome project the assembled the ....billion bp of DNA, ……3.12

64. The _____ is a function of both the number of different species in the community (species richness) and their relative abundances ……Species diversity

65. dinoflagellate belong to ……Protozoans

66. larvae of Echinoderms are ……bilaterally symmetrical

67. contribution of an individual's genotype to the next generation ……Fitness

68. ____are the plants found in a particular area ……Flora

69. Large scale areas of similar vegetation and climatic characteristics are known as .Biome

70. The study of transcriptomes is part of .....genomics…… functional

71. Trochophore larvae develop, which are remarkably similar to the ……Mollusks

72. Features that have no apparent function…… Vestigial Structures

73. ..... is the ordered division and naming of organisms ……Taxonomy

74. Some species known as _____ have unusually strong impacts on community structure, preserving the balance of the community or even making its existence possible…….. keystone

75. Humans have been evolved about .... Year ago…… 5 million

76. Seeds can be maintained for decades or even centuries if the conditions are controlled…… <5% humidity

77. Crabs have ..... Body symmetry ……Bilateral

78. Breeding in annelids occurs in…….. spring

79. _____may also be used to determine the amino acid sequence of a protein……Mass spectrometry

80. In bioinformatics HMM stand for…… Hidden Markov Models

81. The molecular clock says ..... million years ago the human-chimp line split…… ~5

82. Who is not an invertebrate? ……Amphibians

83. property of a human society in which ecosystems (including humans) are managed such that the conditions supporting present day life on Earth can continue. Is known as ……Sustainability

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84. The more variation, the better the chance that at least some of the individuals will have an ..... that is suited for the new environment..... **allelic variant**

85. Pharyngeal gill slits .....**are unique chordate characteristic**

86. Which of the following is a typical feature of prokaryotic cell .....**Absence of nucleus**

87. the silver carp, eats..., a key food for many native fish species .....**plankton**

88. Evolutionary fitness is a measure of ..... **Reproductive fitness**

89. The genome size of living organisms ranges across .... orders of magnitude **09**

90. The laborious ..technique has now been replaced mostly by sequencing using mass spectrometry__ .....**Edman degradation**

91. According to the International Union for Conservation of Nature (IUCN), globally about .....of all known species are threatened with extinction..... **one third**

92. Which of the following are characteristics of prokaryotic cell ..... **all of the above**

93. sea squirt is a model and belongs to a group the..... **Urochordata**

94. Organelles that oxidizes amino acids, fatty acids and alcohol known as… **Peroxisome**

95. The velvet worm_____ traditionally seen as a missing link between the .....**arthropods and annelids**

96. Loss of genetic diversity can imply lack of ...and premature extinction.....**Evolution**

97. ..... is the evolutionary history of a species or group of related species .....**Phylogeny**

98. The spread of Homo out of Africa is presumed to have taken place about ...by Homo erectus..... **1.5 MYBP**

99. Coral reefs which are biodiversity hotspots will disappear in .....**20-40 years**

100. 18th century, ... published a system of taxonomy based on resemblances.....**Carolus Linnaeus**

101. — Ancient order inhabits few small islands of new Zealand .....**Rhynchocephalia**

102. Invertebrates Makes up .... of all Animal species **95%**

103. 'The size of puffer fish, ...of the human genome .....**one-eighth**

104. One of the following is not example of positive interaction among the microbes.....**Amensalism**

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105. Gregor Mendel's experiments with plant hybridization led to his establishing……**laws of inheritance**

106. Cockroach has ..... Body symmetry…….. **bilateral**

107. Arabidopsis employs the .... photosynthetic pathway…….. **C3**

108. The stage of parasitic protozoa that actively feed and multiply are frequently called……. **trophozoites**

109. A taxonomic method that uses a short genetic marker in an organism's DNA to identify it as belonging to a particular species……. **Barcoding**

110. ecologists use to describe the composition of a community are species richness and ...diversity ……..**species**

111. which is not a foundation species ……..**green alga**

112. Black carp eat...., limiting their availability for native fish and damaging shellfish populations ……..**mussels**

113. Haldane (1929): gave the ……..**"Hot, dilute soup" theory**

114. Genetic diversity is the variation at the level of……..**both at individual genes, and In a population**

115. ... carp illustrate the potential impact of invasive species ……**Asian**

116. one of the following is the not part of maintaining balance of the ecosystem…… **protecting food crops**

117. levels of studying are…… **Population, Cell/Organism, Organelle, Molecule, Atom**

118. is a foundation species that forms the basis of the kelp forests off the coast of California…….. **brown algae**

119. intermediate disturbance hypothesis suggests that communities with a ... may have greater species diversity than communities…… **intermediate disturbance**

120. Humans are members of the order ……**Primates**

121. Genetic modification of plants is achieved by adding a specific gene or genes to a plant, or by knocking down a gene with...., to produce a desirable phenotype.. **RNAi**

122. Class Polychaeta is the class of Phylum ……**Annelida**

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123. A valid clade is ---, signifying that it consists of the ancestor species and all its descendants…… **Monophyletic**

124. Extra chromosomal self-replicating circular DNA molecules present in bacteria are called;……..**plasmid**.

125. In leeches, salivary glands excrete which prevents the blood from coagulating……**Hirudin**

126. mutations are often lost from a population over subsequent generations……..**Bad or lethal**

127. In which of the following animal sexual reproduction takes place? ….**Cockroach**

128. A taxonomic unit at any level of hierarchy is called a ……..**taxon**

129. Methanogens fall under the domain of: ……..**Archaea**

130. Sharp beak instead of teeth is characteristic of……..**Order Chelonia**

131. ___ Species are species that have been introduced into areas outside their native range and can cause—or have caused—harm in their new area…… **Invasive**

132. Vestimentiferans belong to phylum…….. **Annelida**

133. …. assumes that the tree that requires the fewest evolutionary events (appearances of shared derived characters) is the most likely…….. **Maximum parsimony**

134. Krill is the major food for…….. **whales**

135. Ecologists use to describe the composition of a community are richness and species diversity…….. **Species**

136. Trypanosoma is spread by ……..**tsetse fly**

137. On 26 June 2000, when US president _ J. Craig venter of Celera genomics, and Francis Collins of the national institutes of health jointly announced that a working draft of the human genome had been completed. ……..**B. Clinton**

138. Sea anemone has__ body symmetry. ……**Radial**

139. Who is not a model organism? ……..**honey bee**

140. _____was probably the first hereditary molecule having the ability to copy itself……..**RNA**

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141. Populations of endangered species are small and tend to lose:……..**Genetic diversity**

142. Group of interacting and interbreeding organisms is known as;…… **population**

143. Systematists have proposed the _____ , which recognizes only groups that include a common ancestor and all its descendants. ……..**PhyloCode**

144. Species richness is the number of different species in a particular……..**community**

145. Assigning functions to genes and recognizing _____ sequences in a model genome can greatly benefit from comparison with a set of carefully chosen reference organisms at defined phylogenetic distances. ……..**Promoter**

146. The following are the major activities of plant breeding One of following gives the correct sequence of events in Plant breeding……

Collection of variation. Selection Evaluation, Release

147. The_____ traditionally seen as a missing link between the arthropods and annelids ……..**velvet worm**

148. Centipedes belong to subphylum. ……..**Myriapoda**

149. Which one is an extinct group of Arthropoda? ……**Trilobita**

150. Acritarches are very likely the cysts of……..**algae**

151. _____theory states that groups of similar organisms are descended from a common ancestor…… **Evolutionary**

152. Seeds which are not of agricultural importance yet hold traditional importance; these seeds are kept in seed banks and are still maintained by some gardeners and farmers are Known as;…… **Heirloom**

153. _____ and wildlife can be conserved by setting up areas like Biosphere reserves Wildlife sanctuaries, National parks, etc……**Forest**

154. __is the study of all low molecular weight cellular constituents. Metabolomics

155. If for genetic modification genes of the species or of a crossable plant are used under control of their native promoter, then they are called…….. **cisgenic plants**

156. Aves are _____…….. **both A and B**

157. The locomotory organ of echinoderm are called …….. **Tube feet**

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158. Antibiotic resistance gene rapidly spread within and between bacterial populations by transposons called. .....R plasmid

159. Two million people died in potato famine in .....Ireland

160. The international human genome project initiated by US was completion due in..... 2000

161. The Proterozoic eon alone accounts for ---- of all geologic time.....42.5

162. Wormlike and algae fossils come from _____ million years old rock in china.....700-900

163. Homo sapiens are believed to have originated around how many years ago..... 100,000-200,000

164. Class Squamata has _____ Backbones of .....100-400 vertebrates

165. The ----- synthesize glucose using solar or chemical energy which is broken down through respiration..... Autotrophs

166. By 2 billion years ago free oxygen was appearing in the atmosphere due to the activity of ------ .....Photosynthetic bacteria

167. In protozoa era only ---------- are recognized. .....Bacteria

168. anon mutations are often lost from a population over subsequent generation.....Bad or lethal mutations

169. In the 18 century Carolus Linnaeus published a system of taxonomy based on resemblance. two key features of his system remain useful today two part names of species and hierarchical classification. .....Two part names for species and hierarchical classification

170. In genome evolution gene number and the complexity of an organism.....Are not strongly linked

171. Medicines from nature account for usage by ___ of the world population. 80%

172. By June 2005, no fewer than _____ prokaryotic organisms and 496 eukaryotes were the subject of ongoing genome sequencing projects.....730

173. The international human genome project initiated by the US national institutes of health and the US department of energy, was launched in 1990

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174. Nodes in gene tree relate either to speciation or gene duplication events.....**Duplication**

175. The entire community of plants and animals in a climatic region..... **Biome**

176. The number of organisms whose genome has been sequenced completely and published is now approaching **300**

177. The euchromatin part of the genome was not completed until mid.....**2004**

178. CRF stands for..... **Conditional random field (CRF)**

179. Cladistics method was developed by .....**Willi Hennig**

180. By June 2005, no fewer than _____eukaryotes were the subject of ongoing genome sequencing projects.....**496**

181. The field of biology specializing in storing and analyzing data .....**Bioinformatics**

182. The possibility of working with molecules on the scale of few micrometers is called.....**micro ecology**

183. First question asked when a species is bought for genome sequencing ---------- .....**What is the size of its genome**

184. The study of transcripts that are present at any time in the cell are called.....**transcriptomics**

185. Important measures used to describe community composition are --------- ..... **Species richness and species diversity**

186. Vertebrates makes --------- % of the animals **5%**

187. The number of allele in a sample ---------- ..... **Allelic richness**

188. Mass spectrometry used to determine .....sequence. **Amino acid sequence of protein**

189. Oparin 1924 work ---------- ..... **Primeval soup theory**

190. Quinine is obtained from quina it is used to treat ---------- .....**Malaria**

191. The common extension of genomics are except --------- .....**Vaccinomics**

192. To assemble a genome from a series of sequence fragment require tremendous.....**Computational power**

193. The first genome sequenced completely was that of ---------- ...**Hemophilus influenza**

194. Annelids do not have feeding style ------- .....**Filter feeders**

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195. Modern bread wheat is a mixture of recombined genes from -------- …… **3**

196. They possess notochord in any period of their life -------- …… **Chordata**

197. Which one is not related to human genome project ……**three synteny analysis, phylogenetic, foot printing, phylogenetic shadowing. Other than following**

198. Species differ from foundation species in that they are more like to belong to……**All of above**

199. Which is not related to endosymbiosis ---------- …… **Prokaryotic cell study**

200. Which is not the part of horizontal gene transfer-……**Conjugation, transformation, transduction. Other than these**

201. Which one in not related to protozoa reproduction. ……**Parthenogenesis**

202. Taxonomy was established by --------- . ……**CAROLUS LINNAEUS**

203. Linus Pauling called science a way to find truth or logic , select correct…… **Truth**

204. There are some so called --------- to identify ecologically genes in incompletely genome…… **Pregenomic molecular approaches**

205. The international human genome project initiated by except ……**US national institutes of genetics**

206. Comparative genomics is developing an increasing array of bioinformatics technique except ---------- …… **Synteny shadow**

207. Keystone species differ from foundation specie higher trophic level and top predator ,tend to modify their environment…….. **All give above**

208. Eukaraya have been evolved from since --------- million year ago…… **Two**

209. Types of selection which favor more than two extreme phenotype is……**Disruptive selection**

210. Which of the following is not mode of feeding of annelids ---------- …… **Predators**

211. A philosopher concept of value of something independent of its value to anything else called ------- ……**Intrinsic value**

212. Major technological developments of 1990 computing, communication, micro technology…….. **All of above**

213. ---------- are found in more than one copy in the genome…….. **Paralogues genes**

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214. In Edman degradation technique study of all the proteins in the cell which is not related. ……**Fragment pattern is not compared with genomic sequence**

215. Which is not true about protein degradation. ……**Sequencing using mass spectrometry replaced by degradation**

216. _____ is used to cleave the protein. ……**Trypsin**

217. Arabia thaliana's is.....:…… **Fly**

218. All peptide technique are.....**All**

219. ..... Is used to predict the protein structures……**Mass spectrometry**

220. Trochophore resembles like ..... **Larvae**

221. Plasmid Size ..... **2 to 200 bp**

222. Humans differ from gorilla..... **bipedal**

223. Characteristics That's are transfer into next generation ……**Traits**

224. Kimber Ella is .....**Animal**

225. Coral reesf are disappeared due to ..... **Global Warming**

226. Which institute was not involved in human genome project?.....**U.S.Center of health**

227. Muscular pharynx of annelids is called ..... **Eversible proboscis**

228. Segmentation is confined to rear portion of annelids called ..... **Opisthosome**

229. Birds are ---------- ..... **Homeothermic**

230. Turtles and tortoise belong to order ..... **Chelonian**

231. Biodiversity is the result of 3.5 billion years of ..... **Evolution**

232. 80% of human food supply comes from ------- kinds of plants.....**20**

233. ___is the application of biotechnology and molecular biology.....**Molecular breeding**

234. Edward Wilson described the ..... **Acronym**

235. ---------- of all species on earth will be extinct by 2050.....**30%**

236. .-------- ecosystems are the most threatened ecosystems..... **Freshwater**

237. Over ------- of global fisheries are being overfished.....**25%**

238. Ayubia national park is the first --------- in Pakistan..... **Reserve forest**

239. Effective population size is much smaller than ------ population size. ..... **Census**

240. Seeds can be maintained in seed bank at ---------- .....**20°C**

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241. CITES the convention in international trade in endangered species set up in …….**1988**

242. Conjugation, transformation and transduction are_____ gene transfer…. **Horizontal**

243. Oparian 1924 gave the_____.....**Primeval soup theory**

244. RNA shows enzymic properties called …….. **Ribozymes**

245. Analogies are also known as ……**Homoplasis**

246. ---------- years ago free oxygen was in atmosphere…….. **2 billion years**

247. Human order primates consist of -------- species……**180**

248. ---------- groups of organisms by common descent……**Cladistics**

249. All known protozoic animals were……**Softed bodied**

250. Malaria is caused by plasmodium technically a member of phylum …….**Sporozoa**

251. Lyme disease caused by …… **Tick**

252. Larva of annelids trochophore similar to the…… **Mollusks**

253. Gut tissues form an organ …… **Trophosome**

254. Trophosome play a role in…….. **Nutrition**

255. ---------- is a small ball above the tube foot……..**Ampullae**

256. Vertebrates are evolved from ---------- 540 million years ……**chordates**

257. Chordates are bilaterian animals that belong to clad of animals known as……..**Deuterostomia**

258. The study of traditional reptile orders …… **Herpetology**

259. Humans use --------- species for food clothing and shelter……**40,000**

260. ---------- of all mammals will be extinct in 20 years. ……..**20%**

261. Homology is similarity due to ---------- b/w wolf and coyote……**Shared ancestry**

262. Analogy is similarity due to --------- in similar condition…… **Convergent evolution**

263. A ---------- is a group of species that includes on ancestral species……**Clade**

264. ---------- an animal from latest Proterozoic rocks in Russia…… **Kimberella**

265. Three types of homology, 1-structural 2-developmental 3-genetic ……**All**

266. Homologues structures ---------- ……**Same structure & different functions**

267. Plant breeding is a practice estimated to Date Back:…… **9.000 to 11.000 years**

268. Family Siboglinidae has _____ species……**120**

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269. Chinoderms have sticky and toxic tenticles which are shot out the anus known as:…….. ……**Cuvierian tubules**

270. Interacting populations which significantly affect each other's distributions and abundance (intertidal, hot spring, wetland) is known as:……**Community**

271. Alligators fall under which of the following order?.....**Crocodila**

272. The odd creature known as Kimberella has been found from the _____ of Russia……..**latest Proterozoic**

273. _____ is not an important factor that influence community structure……**Reefs**

274. The plants resulting from adding a gene are often referred to as;……**Transgenic**

275. Which of the following is no available in animal cells?.....**Plastids**

276. A supportive rod that extends dorsally in the animals to the body cavity into the tail is called;…… **Notochord**

277. [How much amount of human food supply comes from plants?.....**80%**

278. Bilateral symmetry is seen in the body organization of,……**Annelids. arthropods and vertebrates**

279. Humans are member of the family:…….. **Hominidae**

280. Some tropical rainforest communities have _____ numbers of species packed into each cubic meter……..**Huge**

281. Bipinnaria larva is the characteristic of:…….. **Echinoderms**

282. In vitro early earth conditions were recreated by:…….. **Miller and Urey**

283. In some cases the primary structure of a gene product (a protein) may be the starting point of gene……..**Discovery**

284. In leeches salivary glands excrete _____ which prevent the blood from coagulating……**Heparin**

285. The international human genome project initiated by the US national institutes of health and the US department of energy. was completion due in……**2005**

286. The two-part scientific name of a species is called a;……**Binomial**

287. A _____ grouping consists of an ancestral species and some. but not all. of the descendants…….. **Paraphyletic**

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288. Sinorhizobium meliloti is a plant,……..**Bacteria**

289. Microlesions are:…… **Both a & b (Point mutations & Frame shift mutations)**

290. Physical location of a gene is known as;……**Locus**

291. Protochordates are divided into:…….**Both a & b (Urochordates & 'Cephalochordates)**

292. The highest number of mammal species _____. per square kilometer. occurs in the Amazon region of South America……**179-228**

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